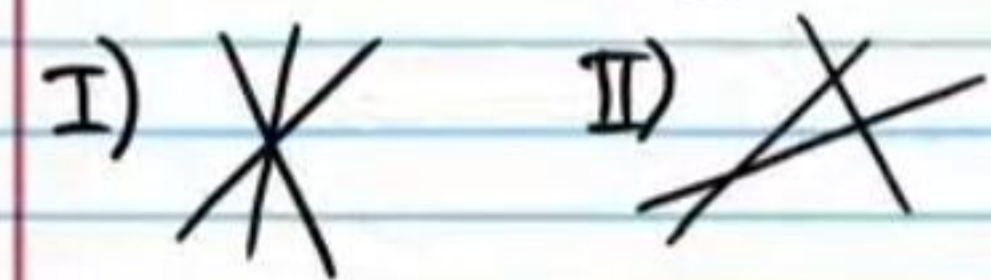
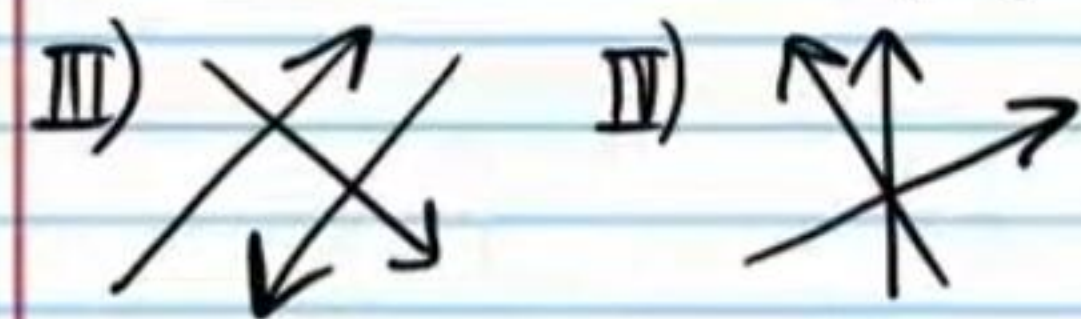
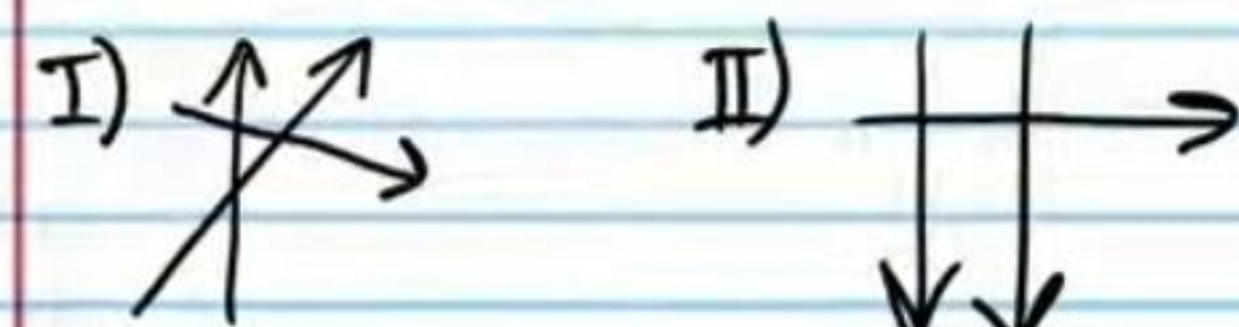


Points of Concurrency Generated by Triangles (Homework)

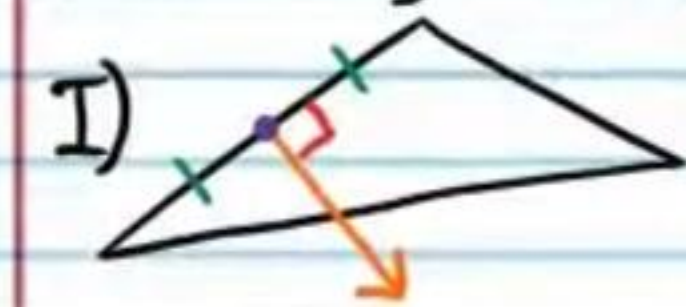
① Which segments are concurrent? You may select more than one group.



② Which rays are concurrent? You may select more than one group.



③ Identify the line, ray or segment as a perpendicular bisector (PB), angle bisector (AB), median (M), or line (or ray or segment) containing the altitude (Alt).

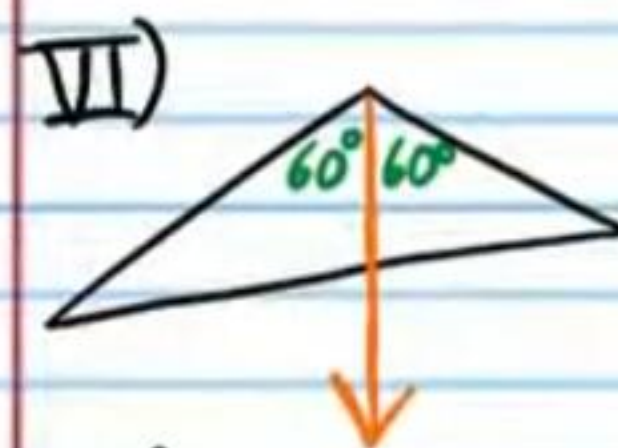
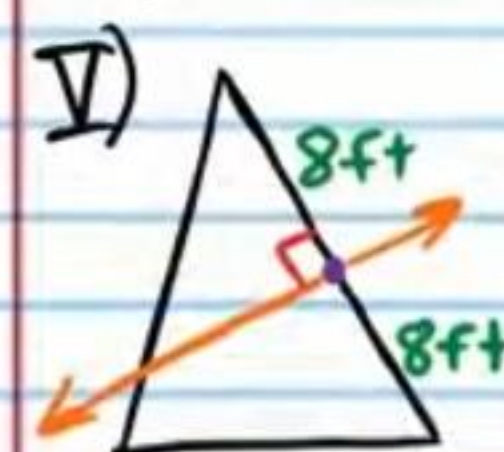
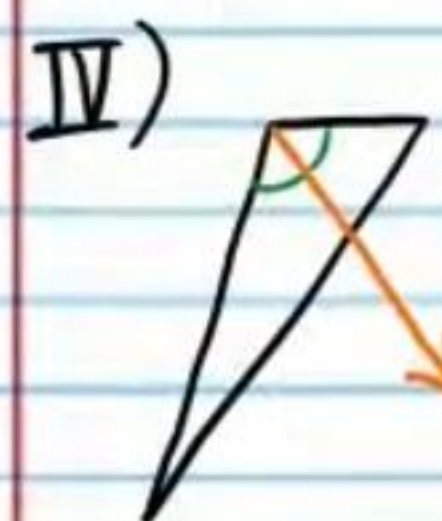
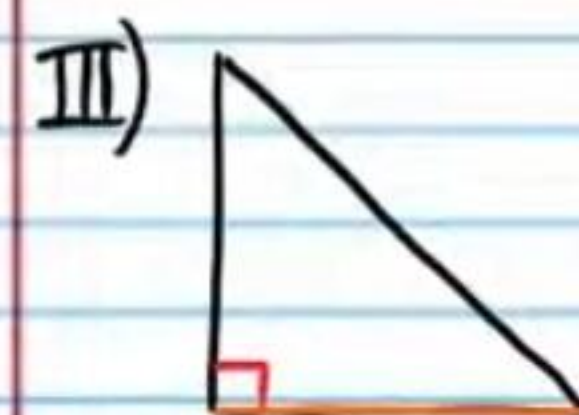
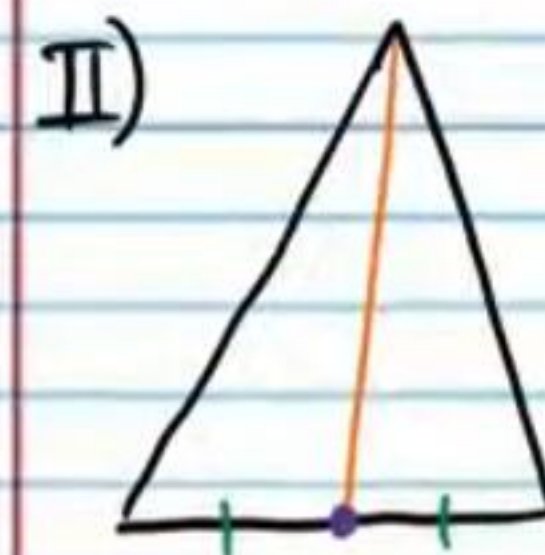


Answers

I and IV

IV

PB



M

Alt

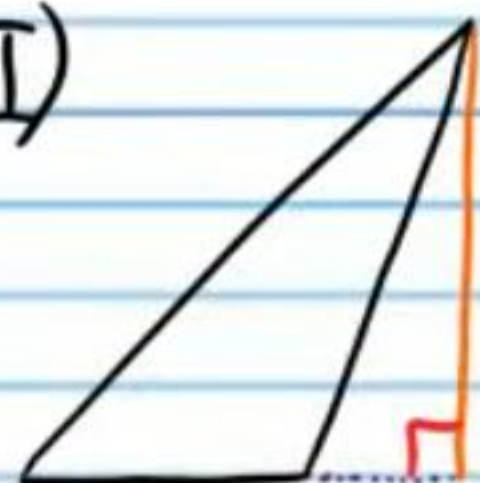
AB

PB

AB

Alt & PB & M

VIII)

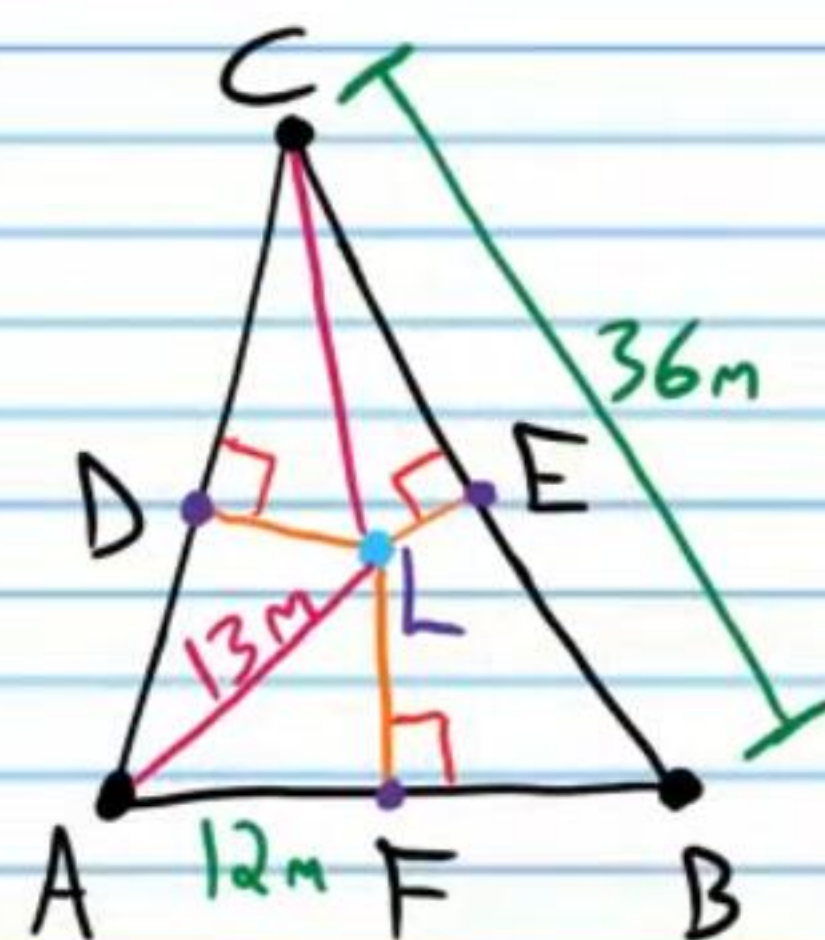


Alt

If point L is the circumcenter of the triangles below, find the quantities listed.

④

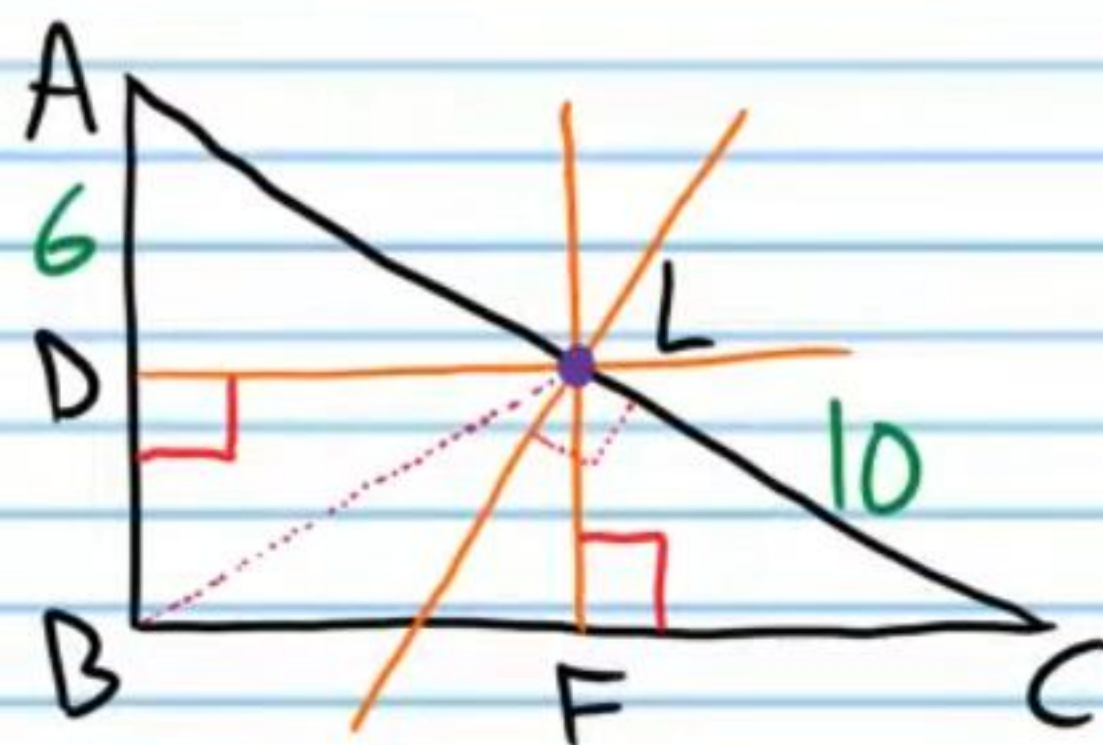
- a) FB
- b) CL
- c) CE
- d) LF



- a) 12m
- b) 13m
- c) 18m
- d) 5m

⑤

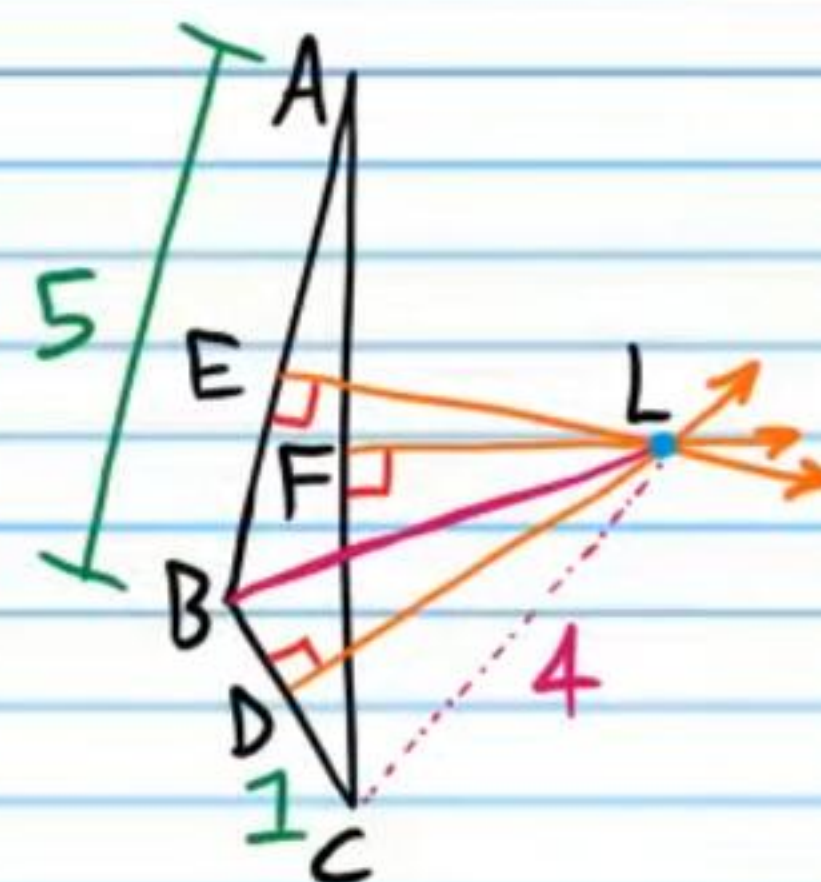
- a) DB
- b) BL
- c) AC
- d) DL



- a) 6
- b) 10
- c) 20
- d) 8

⑥

- a) BC
- b) BL
- c) AE
- d) DL

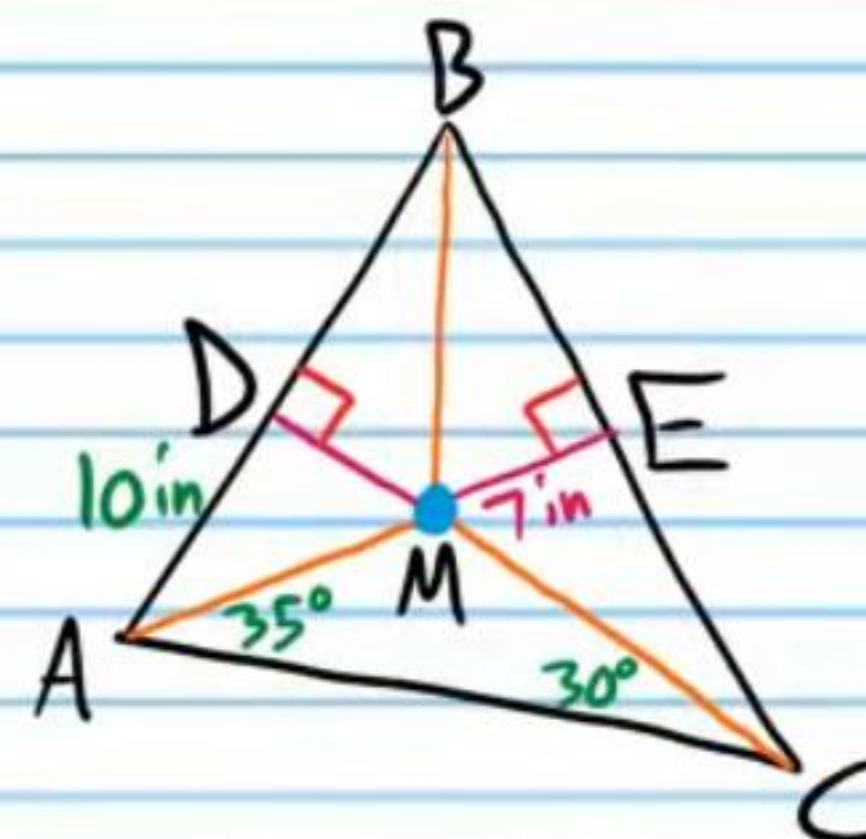


- a) 2
- b) 4
- c) 2.5
- d) $\sqrt{15}$

If point M is the incenter of the triangles below, find the quantities listed.

⑦

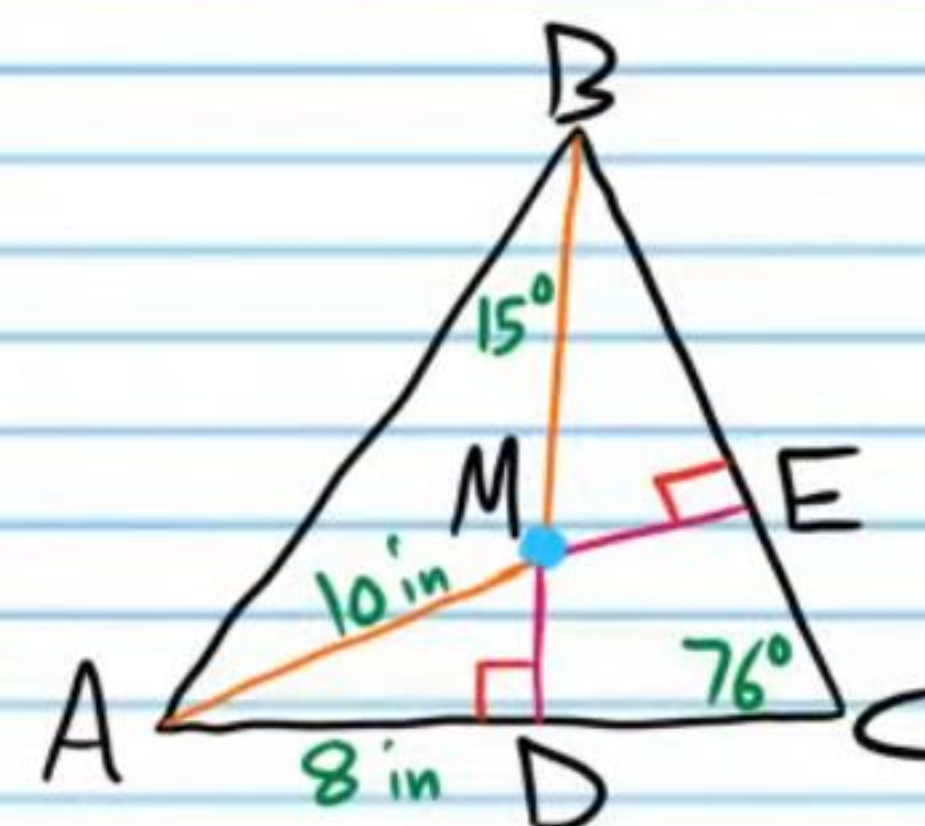
- a) $m\angle MAD$
- b) DM
- c) $m\angle ABC$
- d) AM



- a) 35°
- b) 7 in
- c) 50°
- d) $\sqrt{149}$ in

⑧

- a) $m\angle DBC$
- b) $m\angle MAB$
- c) $m\angle AMD$
- d) ME



- a) 15°
- b) 37°
- c) 53°
- d) 6 in

If the segments below are the medians of the triangles, find the quantities listed.

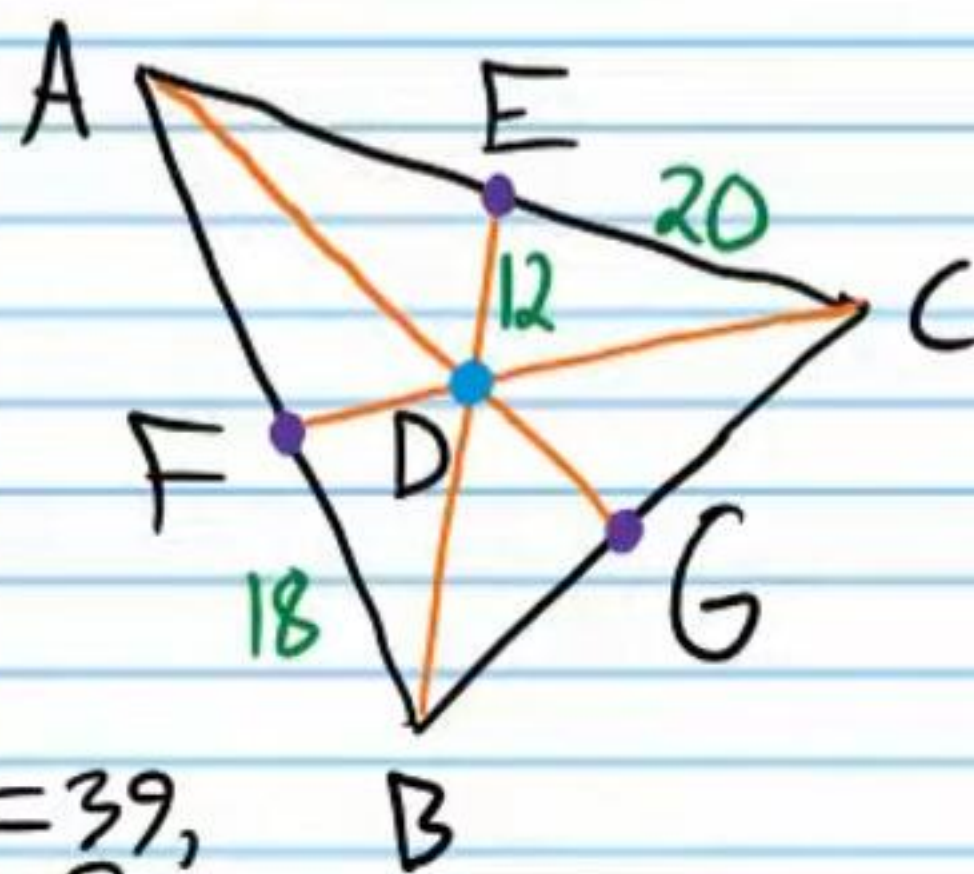
⑨

a) AF

b) AC

c) DB

d) If $AG = 39$, then $DG = ?$



a) 18

b) 40

c) 24

d) 13

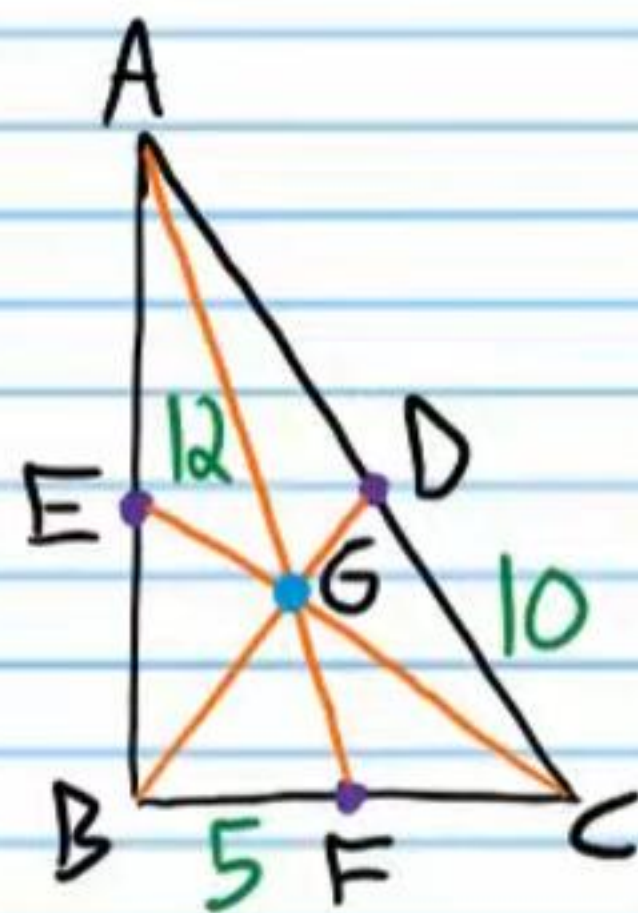
⑩

a) BC

b) AD

c) AF

d) If $BG = 6$, then $GD = ?$



a) 10

b) 10

c) 18

d) 3

If point D is the orthocenter of each triangle below, find the quantities listed.

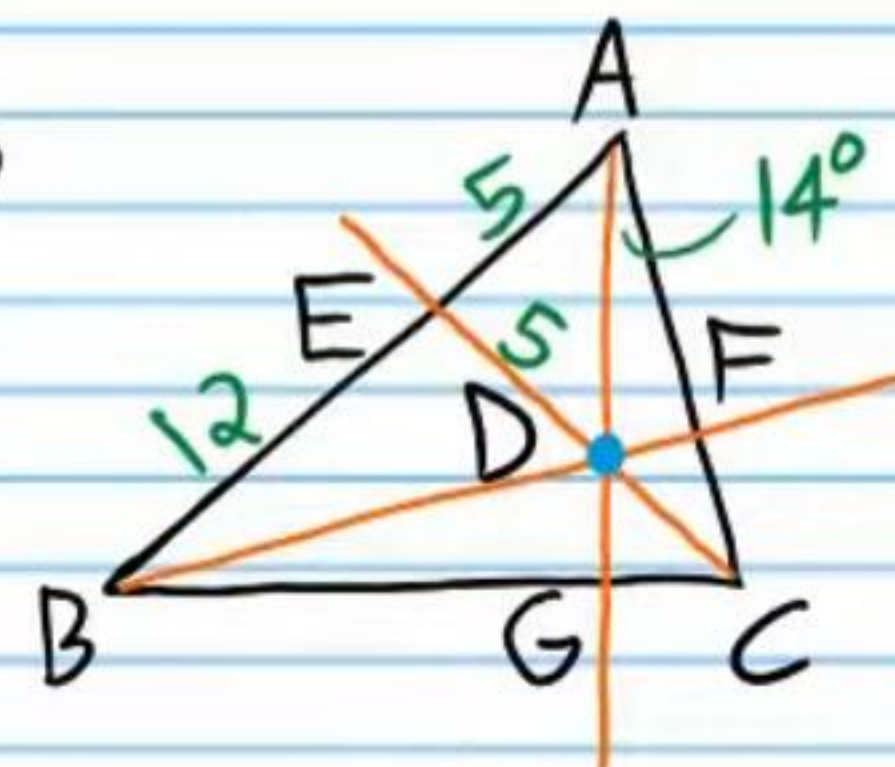
⑪

a) $m\angle AED$

b) $m\angle BCA$

c) BD

d) AD



a) 90°

b) 76°

c) 13

d) $5\sqrt{2}$

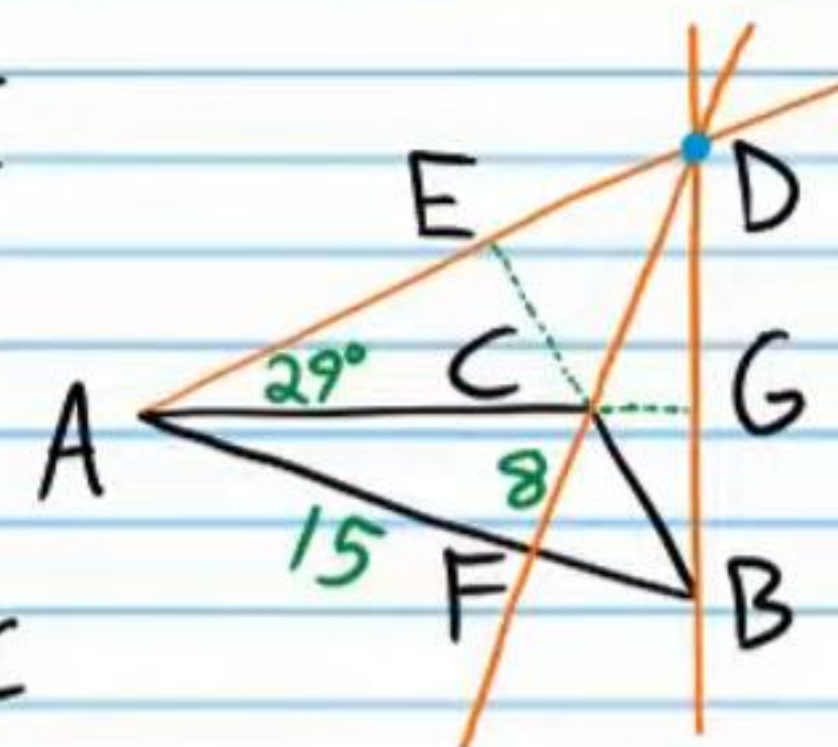
⑫

a) $m\angle ACE$

b) $m\angle AFC$

c) AC

d) $m\angle DGC$



a) 61°

b) 90°

c) 17

d) 90°

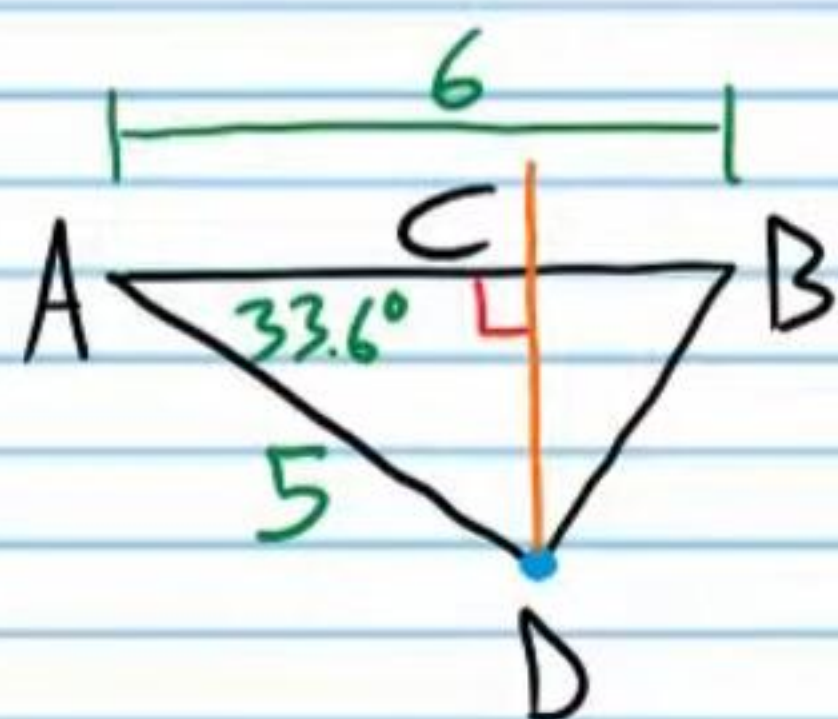
⑬

a) DB

b) $m\angle ADB$

c) $m\angle ABD$

d) $m\angle CDB$



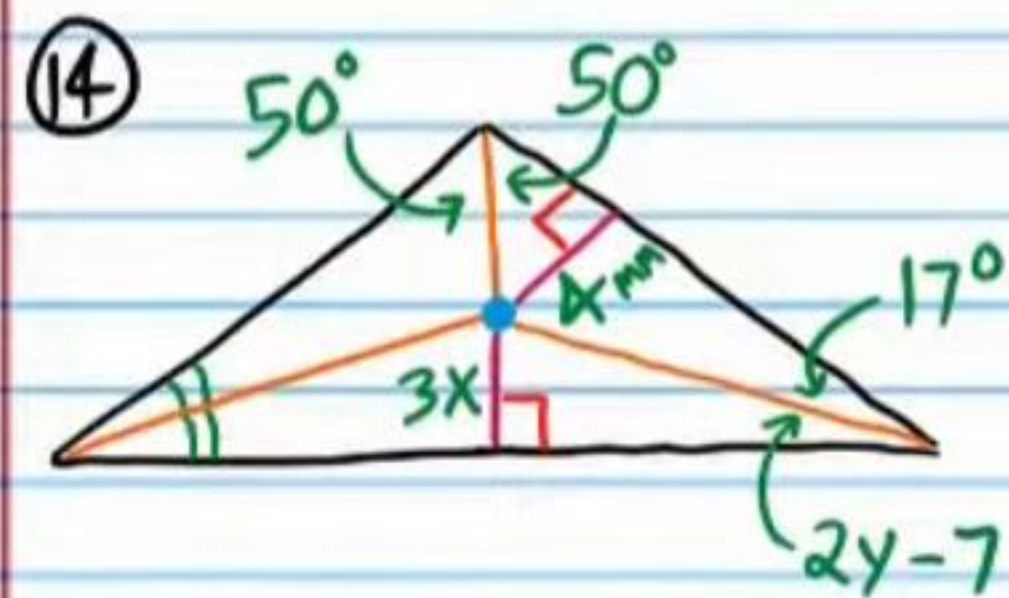
a) $\sqrt{11}$

b) 90°

c) 56.4°

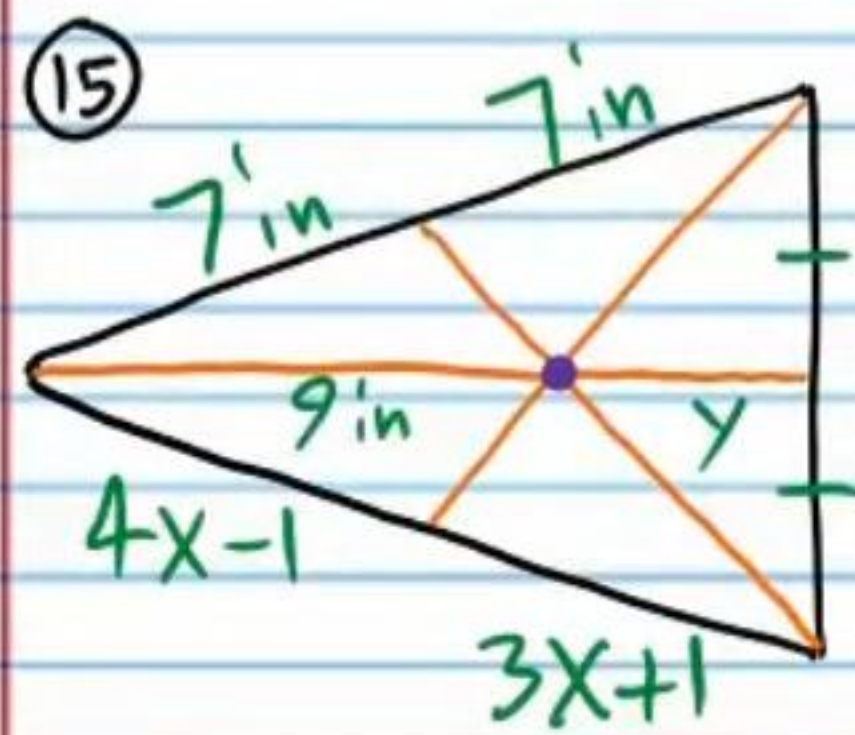
d) 33.6°

Find x and y .



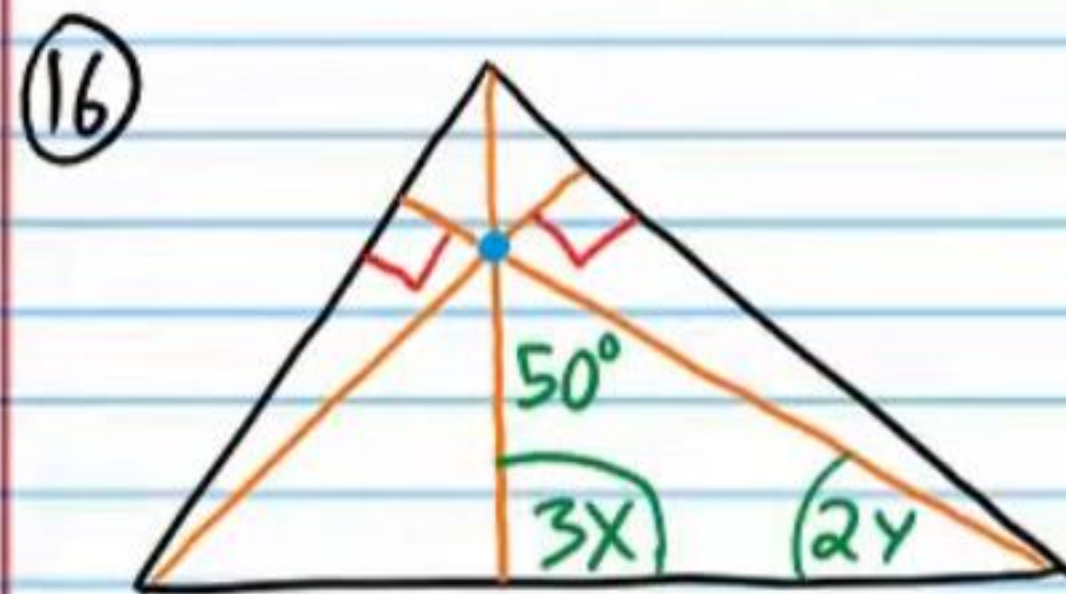
$$x = \frac{4}{3}$$

$$y = 12$$



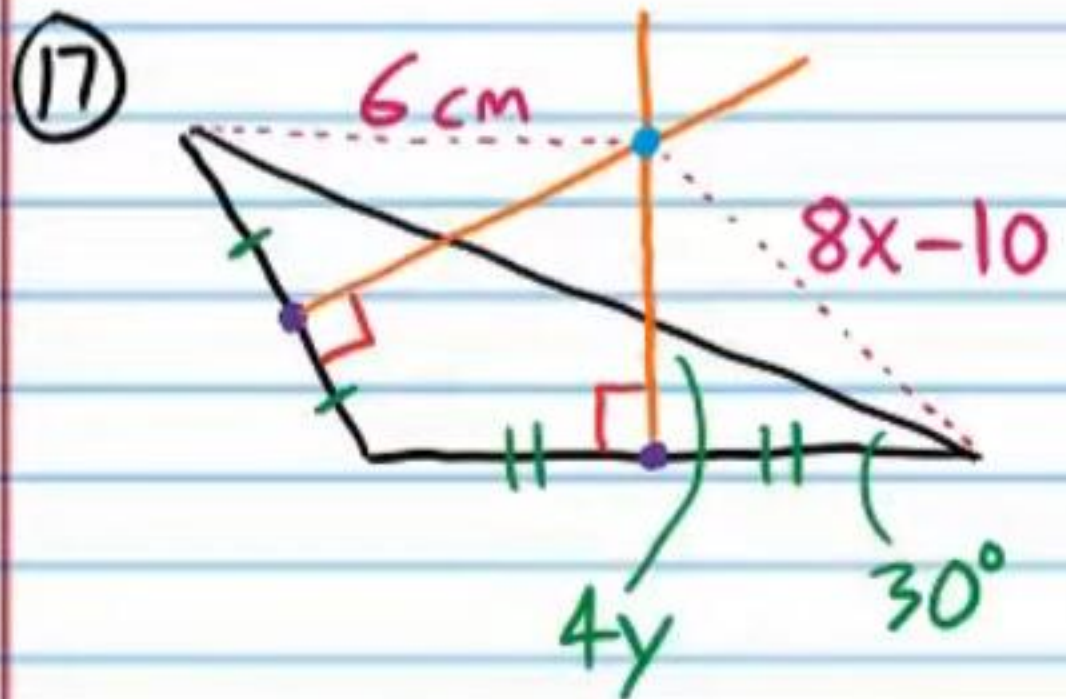
$$x = 2$$

$$y = 4.5 \text{ in}$$



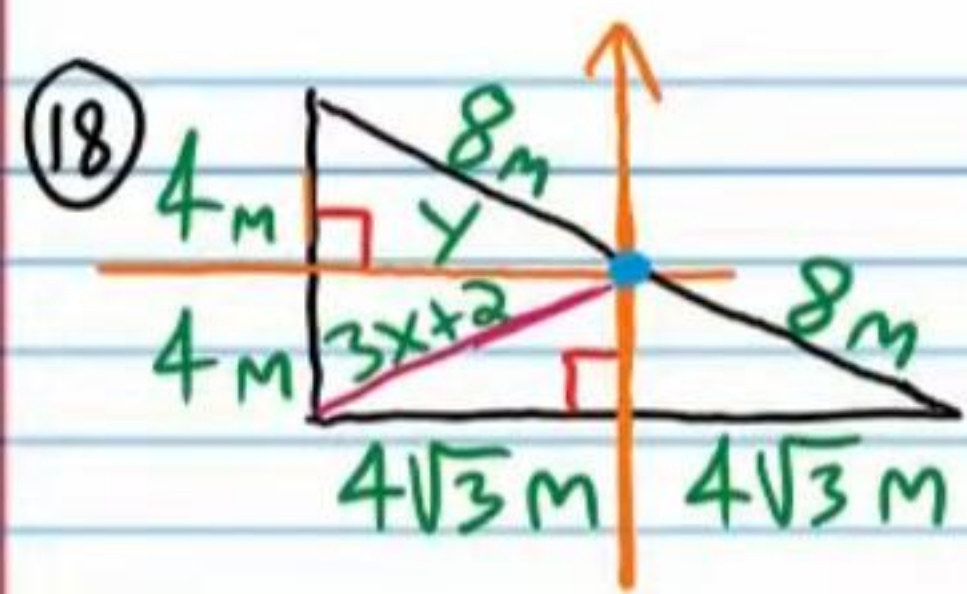
$$x = 30$$

$$y = 20$$



$$x = 2$$

$$y = 15$$



$$x = 2$$

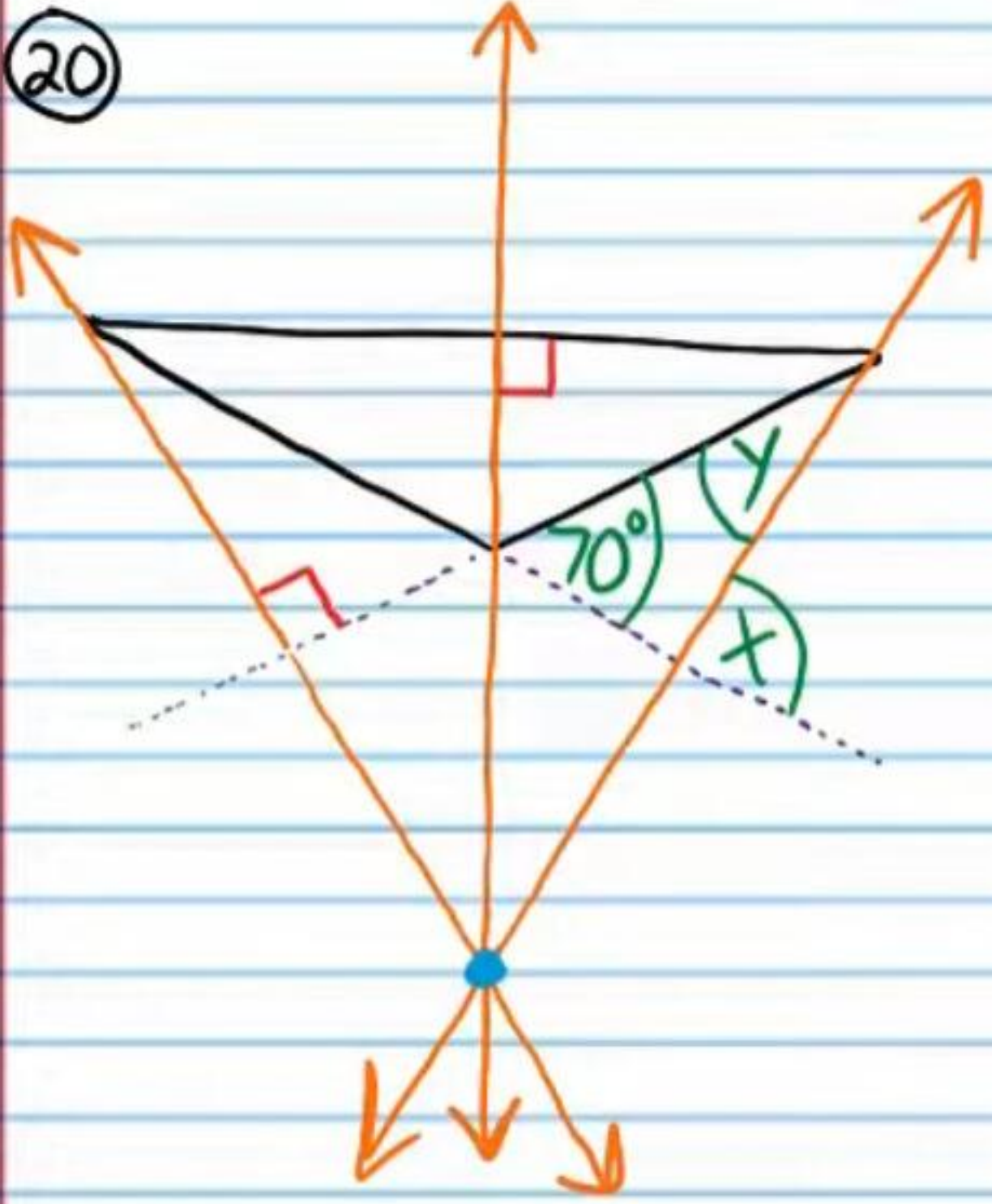
$$y = 4\sqrt{3} \text{ m}$$

19 Hint: The overall triangle must be a right triangle.



$$x = 15^\circ$$

$$y = 5\sqrt{2} \text{ ft}$$



$$x = 90^\circ$$

$$y = 20^\circ$$